

Incorporating production-side variability into nitrogen footprints: linking process models with a life-cycle-oriented assessment framework

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KEY CLAIM

Production-informed nitrogen footprints can connect farm-level nitrogen management to consumer and policy action.

BRIEF JUSTIFICATION, REMAINING CHALLENGES

Nitrogen footprint approaches have been effective in linking food consumption with reactive nitrogen losses along supply chains (Leach et al., 2012). However, representative virtual nitrogen factors (VNFs) can mask meaningful production-side differences within the same food category. Previous work has shown that VNF estimates can depend on the mix of production systems, as demonstrated for seafood (Oita et al., 2016), and can respond to changes in nitrogen use efficiency (NUE) and nitrogen recycling across countries and over time (Oita et al., 2020). Yet such production-side heterogeneity has generally been incorporated through weighted averaging into representative VNFs, rather than retained as differences among specific production conditions.

Our current work asks how much more of this production-side signal can usefully remain visible in nitrogen footprint assessment. In Japan, Ogino et al. (2026) developed a watershed-scale VNF framework coupled with SWAT that explicitly accounts for net nitrogen supply from, and accumulation in, soil nitrogen stocks and disaggregates cultivation-stage nitrogen losses by pathway. Coupling the framework with spatially distributed SWAT outputs allows subnational variation to emerge in crop-specific VNF estimates and helps interpret the production processes underlying those differences. In Germany, ongoing spatially resolved work examines how management practices and field conditions can be reflected in nitrogen footprint assessment. Together, these lines of work suggest that more detailed production information can broaden the representation of production-stage nitrogen flows and distinguish differences within the same crop category under different management and environmental conditions, rather than averaging them all into a single representative VNF.

The challenge is to introduce decision-relevant differentiation in VNFs within the same food category without making nitrogen footprint assessment unnecessarily data-intensive, uncertain, or difficult to communicate. The appropriate level of detail will depend on the magnitude of production-side variability, the information available, and the intended use of the footprint. Detailed process-based assessments may not be feasible or necessary in every case, so it is important to determine when simpler production information is sufficient and when information on specific loss pathways or soil nitrogen stocks adds meaningful value. Practical use also requires credible ways to retain, aggregate, and communicate differentiated production information along supply chains while preserving comparability and avoiding false precision.

POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

Production-informed nitrogen footprints can retain production-side differences that are otherwise averaged out. When farm-level NUE adequately captures differences among management practices, it could be incorporated into production-informed nitrogen footprint calculations to link these differences to downstream stages of the food supply chain, providing a simpler way to reflect production conditions than detailed process-based modelling. Linking nitrogen footprint information to loss flows in regional and national nitrogen budgets can help relate those losses to the food products and demand associated with them. Conversely, nitrogen budgets can place product- and management-specific footprint differences within the broader context of nitrogen inputs, losses, transfers, and stock changes. Together, these linkages could connect where and why nitrogen losses occur with the products and demand associated with those losses, providing a stronger basis for policy, procurement, and retailer and consumer support for lower-N-loss production.

QUESTIONS TO SEED THE DISCUSSION

1. When does production-side variability justify differentiated virtual nitrogen factors rather than relying on food-category averages?
2. How can production-informed virtual nitrogen factors from detailed assessments be connected with farm-level NUE and regional nitrogen budgets for broader practical use?

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